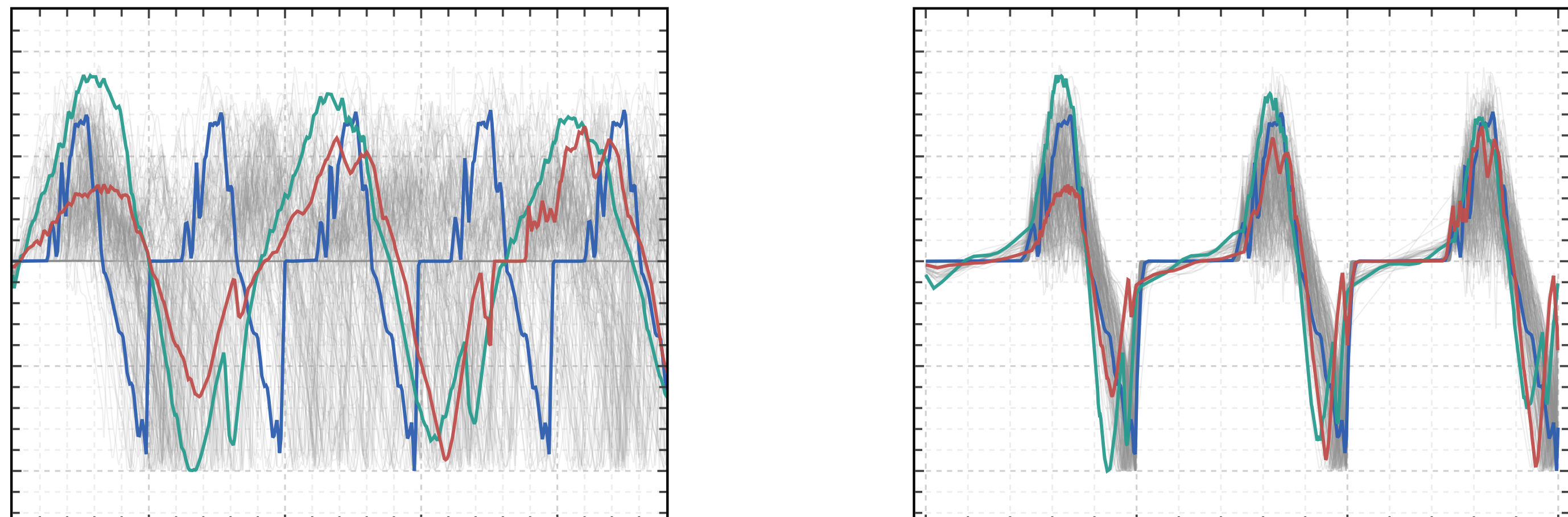


WANTED: SIMULATIONS IN NEED OF A SURROGATE

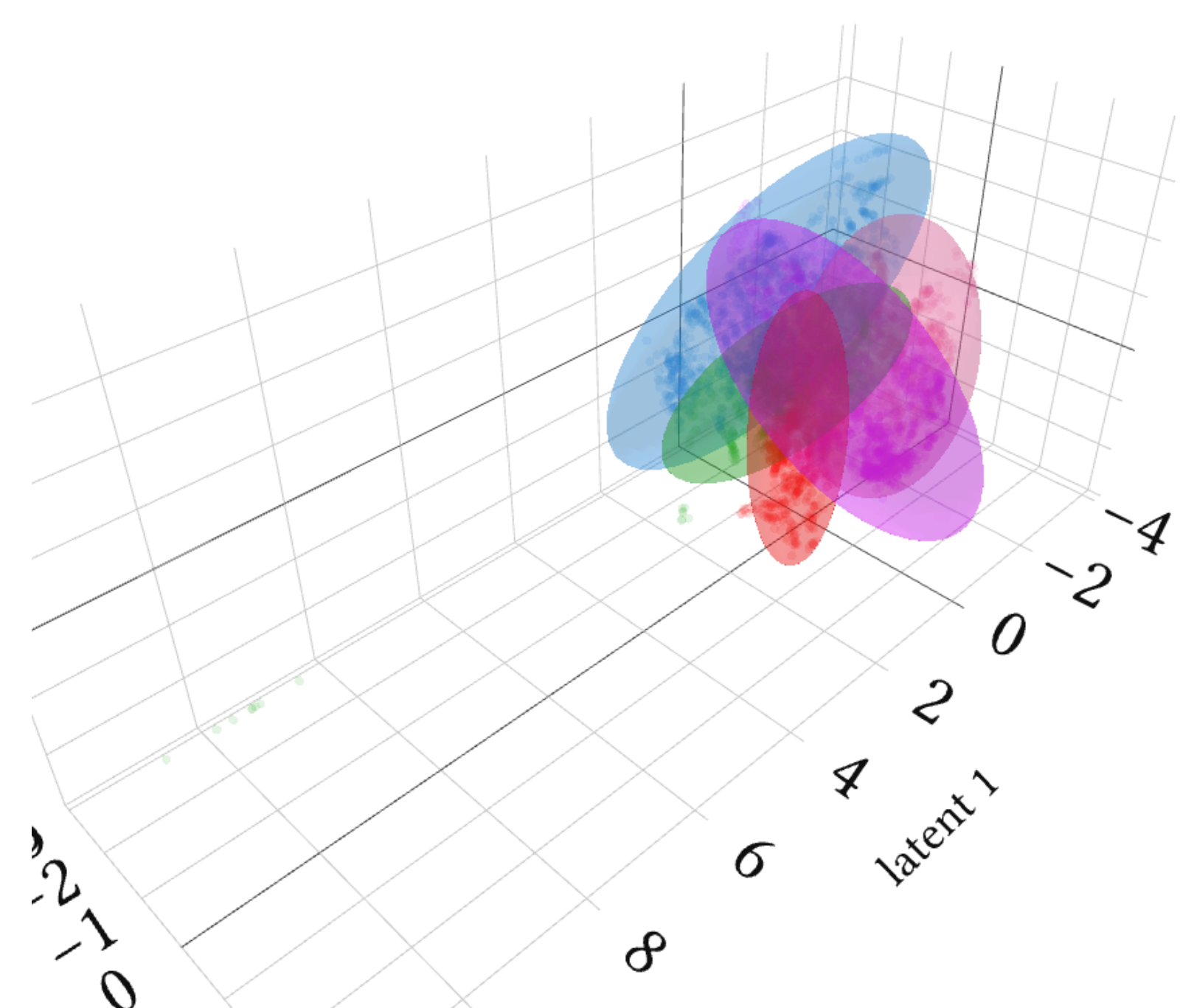
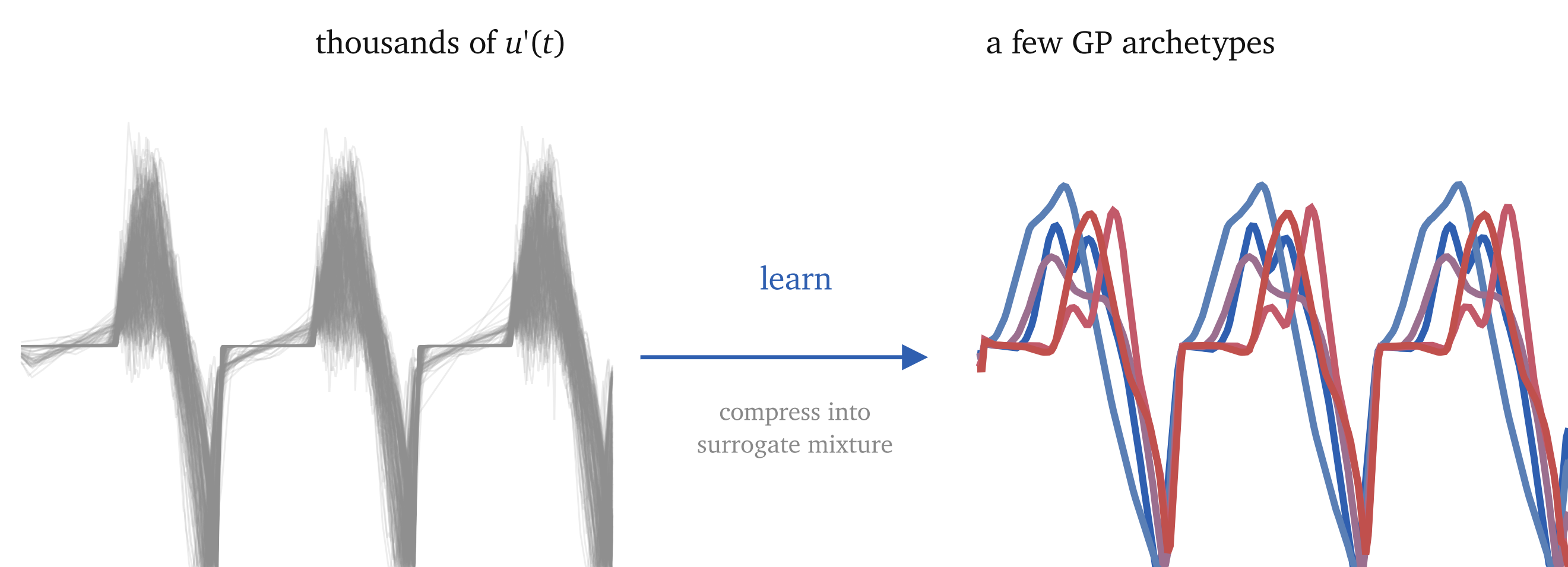
THOUSANDS OF SLOW SIMULATIONS...

High-fidelity vocal fold simulations: hours of compute, one airflow cycle each.



...BECOME THREE NUMBERS AND ERROR BARS

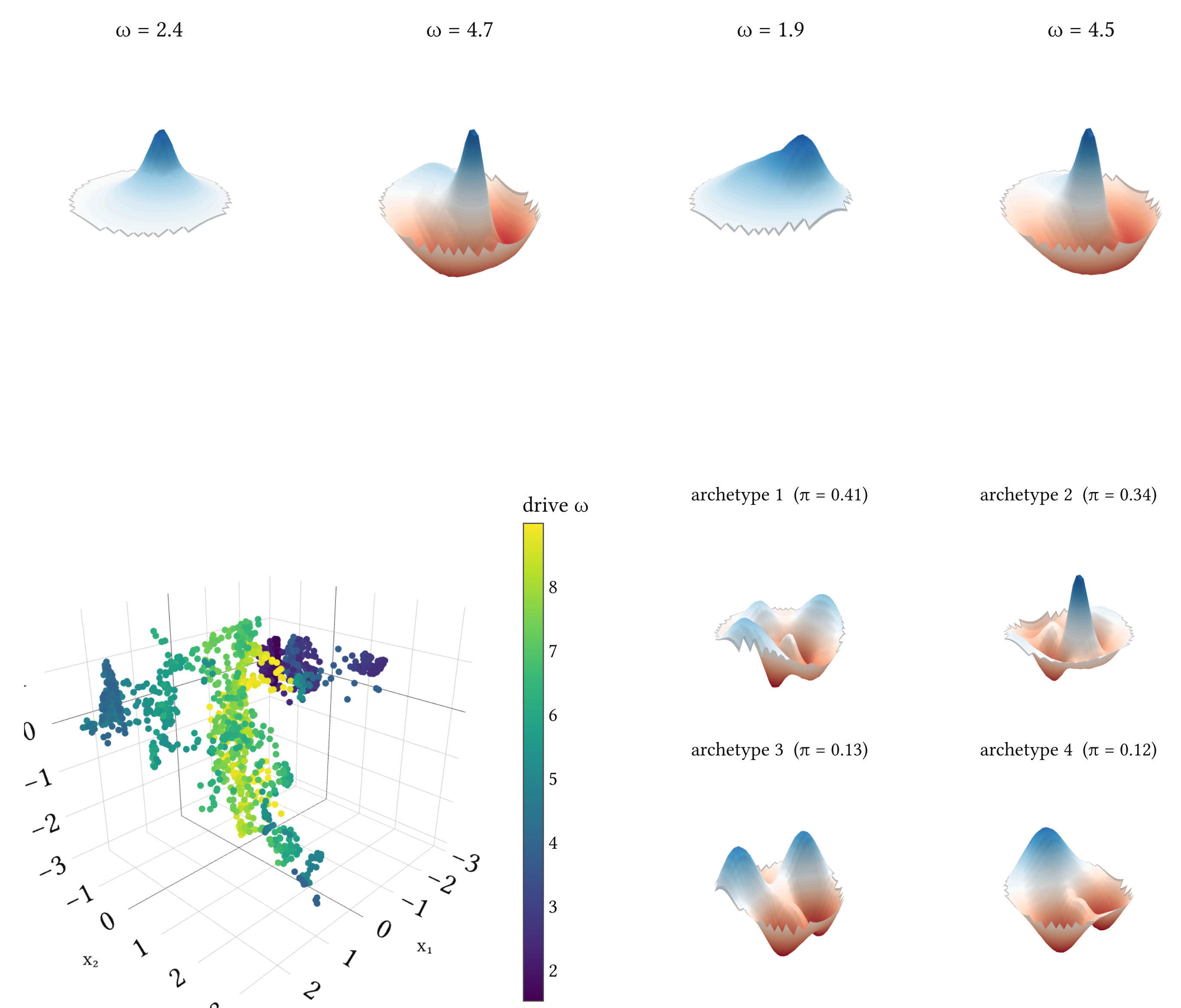
Every simulation becomes a point on a 3-D map with its own error bar. A handful of archetypes explains them all.



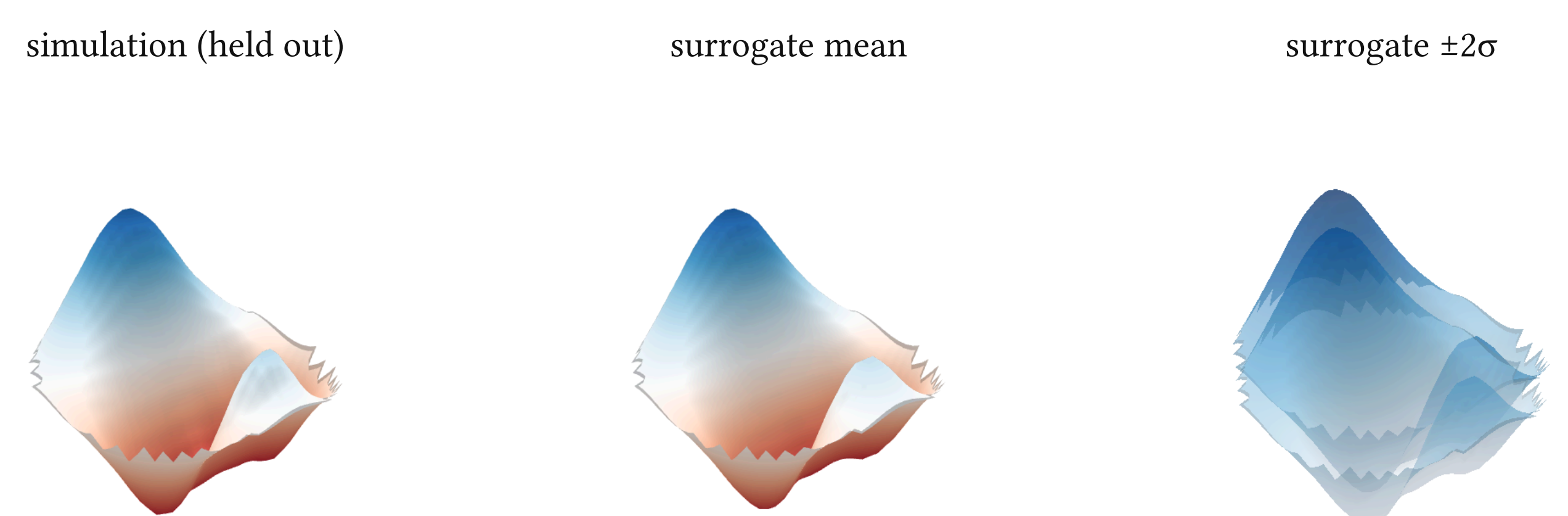
Latent map of 6,500 simulated airflow cycles, one ellipse each.

IT IS NOT ABOUT SPEECH

Side quest for this poster: 1,600 point-driven membrane simulations. Same pipeline, untouched.



Latent map coloured by drive frequency, and the four archetypes it learned.



A held-out simulation, its reconstruction from three numbers (8% error), and the $\pm 2\sigma$ band.

YOUR SIMULATOR HERE

Finite-element models, vibrometry, parameter sweeps you cannot afford to redo. Bring them.

